When On-Chain Flow Signals Work and When They Don't

AMM-Layer Stablecoin Stress Propagation Across Five Episodes, a Provenance-Gated Analysis of Curve Finance TokenExchange Flow

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Abstract

Empirical contagion studies routinely conflate genuine *contagion*, a crisis-driven shift in linkage, with constant *interdependence*. Using Tier-A on-chain Curve `TokenExchange` flow across five stablecoin stress episodes, we show the distinction is empirically sharp and, decisively, *mechanism-specific*. For the June 2023 USDT/Curve episode, an *endogenous* pool-imbalance shock, cross-pool flow coupling rises from near-zero in calm ($\hat{\rho} = 0.09$) to $\hat{\rho} = 0.53$ during the acute window (a *contemporaneous* lag-0 co-movement, not early warning; Fisher $z = +2.82$, block-bootstrap one-sided $p = 0.035$), whereas none of the four *exogenous* episodes (Terra/LUNA, USDC/SVB, FTX, BUSD) shows this activation despite identical Tier-A data, so the null is mechanism-specific, not a data gap. The same split governs structure and detection. On-chain arbitrage is stabilizing in calm markets but flips to amplifying under endogenous stress, price discovery is on-chain for DeFi-native shocks (the Curve pool deviates 37× more than Binance) but CEX-led for bank-run shocks, and an online Hidden Markov Model run causally on on-chain state reaches AUROC 0.95 on Terra/LUNA (versus 0.50 on market data), raising the alarm 116 hours earlier. The operational implication is concrete. DeFi stress surveillance needs *separate monitoring architectures* for endogenous versus exchange-originating shocks, and a provenance-aware three-gate pipeline underpins every claim.

CCS Concepts

• **Applied computing** → **Economics**; • **Computing methodologies** → *Machine learning*.

Keywords

stablecoin, AMM, Curve Finance, DeFi stress, provenance, claim gate, contagion networks, online detection, hidden Markov model

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1 Introduction

On 15 June 2023, the Curve Finance stablecoin pool recorded a 12.5% deviation of USDT from its one-dollar peg while Binance simultaneously showed 0.3%. The pool was 37× **more sensitive** to the stress than the exchange. A risk manager monitoring only exchange prices would have seen almost nothing. Thirteen months earlier, during the Terra/LUNA algorithmic collapse, on-chain pool-state data would have raised the alarm 116 hours before exchange-based signals detected the crisis. These two numbers, 37× more amplitude on USDT/Curve and a 116-hour detection lead on Terra/LUNA, define the surveillance gap this paper measures.

Four distinct stablecoin systems experienced significant stress across these episodes. The algorithmic UST/Terra collapsed in May 2022. FTX's implosion triggered exchange runs in November 2022. USDC briefly de-pegged following Silicon Valley Bank's failure in March 2023. And Curve Finance pools became severely imbalanced in June 2023. *Curve Finance* is an *automated market maker* (AMM), a decentralised exchange whose prices are set not by matching buyers and sellers, but by a mathematical formula (StableSwap [4]) applied to the ratio of token reserves held in a liquidity pool. When one stablecoin in the pool is in excess demand, the pool's balance tilts, its price deviates from the reference exchange, and arbitrageurs flow in or out to restore parity, generating the on-chain flow signals we study. Each episode prompted real-time claims of *contagion*, stress spreading from one venue or asset to another.

Most empirical analyses of these events rely on price data alone and treat all data sources as equally reliable. This paper addresses both limitations by introducing a provenance-aware framework that makes data quality an explicit input to every empirical claim, and then holding every result, including a supervised prediction benchmark, to the same evidentiary gate. That discipline is what lets us separate the findings that survive from the one that does not. The same gate certifies three structural results on Tier-A on-chain flow and rejects a predictive lift that fails to replicate out-of-event.

The surveillance gap. A price-based DeFi stress monitor observes only stablecoin basis widening, an aggregate market outcome, and is blind to the flow-level mechanism beneath it. Curve Finance `TokenExchange` logs, by contrast, record every swap with an exact block timestamp and exact token amount, freely available and updated every 12 seconds. This is a direct observation of the trading flow that *produces* the price move, not a derived signal. Our analysis shows that the cross-pool flow coupling during the June 2023 USDT/Curve episode is strong but *contemporaneous* with the price stress rather than leading it (Sections 5 and 8). The contribution is therefore a mechanism-level decomposition of how stress moves through the AMM layer, not an early-warning claim. No existing stablecoin contagion study uses TokenExchange flow as its primary evidence layer.

Contributions. *Methodologically*, we introduce a provenance-aware three-gate pipeline (data-quality tier → feature tier → statistical significance) and a claim-level taxonomy (A/A DEX-flow, A/A settlement, A/B directional, B/B contextual) that caps every cross-venue claim at its lowest tier and blocks fixture data from reaching any conclusion. This discipline is itself a financial contribution. It provides a replicable framework for auditing model claims before they inform surveillance and risk systems. *Empirically*, on five stress episodes we report four results. (1) regime-switching contagion, USDT/Curve cross-pool coupling rises from $\hat{\rho} = 0.09$ (calm) to 0.53 (panic), Fisher $z = 2.82$, resting not on one statistic but on three *converging* methods on the same episode (Forbes–Rigobon, 96.5% of bootstraps positive. An online HMM detecting the regime at AUROC 0.93. A TVP-VAR placing 94% of spillover post-onset), present only for the endogenous shock despite all five events carrying Tier-A A/A pairs. (2) arbitrage flipping from stabilizing to amplifying under stress ($z = +3.84$ USDT/Curve, +3.69 BUSD, −2.80 FTX). (3) on-chain price discovery, the Curve pool leading the exchange and deviating 37× more for DeFi-native stress. And (4) a machine-learning result on which evidence layer detects stress first, an unsupervised online detector run causally on Tier-A on-chain state recovers endogenous DeFi-native stress that the Tier-B market layer registers at chance (Terra/LUNA causal AUROC 0.95 vs 0.50, 116h earlier), while market data is required for exchange-borne shocks. The layers are complementary, and supervised cross-event prediction fails under concept shift.

Implications for DeFi risk practitioners. Risk managers on CEX-only feeds would be structurally blind to pool-imbalance shocks. Terra/LUNA went undetected 116 hours longer than on-chain monitoring would have allowed. Section 9 gives a replicable decision framework. Endogenous shocks require on-chain detection, exogenous shocks require market signals. Both layers are necessary.

2 Related Work

Stablecoin stress and AMM microstructure. Prior work studies stablecoin stability through runs, reserve adequacy, and safe-asset economics [3, 8, 12], and AMM microstructure through loss-versus-rebalancing and constant-function pricing [2, 13]. These studies establish the economic mechanisms but rely on market-price data. They do not exploit on-chain flow logs as a *direct, execution-grade* observation of the trading that produces the price move. Egorov [4] describes Curve's StableSwap invariant that governs pool pricing. We treat the resulting `TokenExchange` logs as tick-data econometrics [1], execution-grade evidence that market proxies can only approximate, and apply them to the contagion question these studies raise but do not settle.

On-chain contagion measurement. Forbes and Rigobon [7] distinguish *contagion* (a genuine crisis-induced rise in cross-market linkage) from *interdependence* (stable background correlation amplified by higher crisis variance), and provide a heteroskedasticity-corrected Fisher z-to-z two-sample test (see Section 5.1). A significant positive z_{Fisher} means linkage *genuinely increased.* We make this operational on Tier-A on-chain flow, explicitly gating claims by data provenance rather than conflating execution-grade logs with market proxies. Recent ICAIF work applies graph neural networks to on-chain link prediction in Uniswap v3 [15] and agent-based models to crypto market manipulation [9], but targets prediction and manipulation rather than the contagion-versus-interdependence distinction we make execution-grade here.

Methodological rigour in financial AI. Recent ICAIF work emphasizes trustworthiness over raw performance [5, 10, 11]. Our provenance gate provides an auditable, claim-blocking discipline for evaluating ML systems before deployment.

3 Framework, Data, and Provenance

3.1 Three-Layer Network

We model the stablecoin stress environment as a three-layer directed network (Figure 1a).

Layer 1, CEX markets. Nodes are trading pairs at centralized exchanges (e.g. USDC-Coinbase, USDT-Binance, USDT-Kraken). Data are public market feeds. 1-minute OHLCV candles, best-bid-offer snapshots, and aggregate trade data from exchange APIs. We assign these nodes **Tier B**. Useful for market context and timing, but not execution-grade. Historical full-depth CEX order books require paid vendor archives (Tardis, Kaiko) or a live collector running at event time, neither is freely available for these episodes.

Layer 2, AMM pools. Nodes are Curve Finance liquidity pools. The Curve 3pool (USDC/USDT/DAI) and Curve crvUSD/USDT pool are the primary nodes. Data are Etherscan `TokenExchange` logs. Every swap recorded on-chain with exact block timestamps, amounts, and immutable provenance. We assign these nodes **Tier A**.

Layer 3, Settlement flows. Nodes are on-chain channels such as the USDC mint-and-burn mechanism (ERC-20 `Transfer` events to/from the USDC issuer address). Data are Etherscan event logs. Tier A.

Edge tier formula. An edge from node i to node j is capped by the weakest link.

$$\text{tier}_{ij} = \min\!\big(\text{tier}_i,\ \text{tier}_j,\ \text{tier}_f\big), \tag{1}$$

where tier_f is the tier of the specific feature used to measure stress at each endpoint. A Tier-A pool with a derived proxy feature (e.g. `Reserve_imbalance`, computed from an approximate pool-size normaliser) is capped at Tier B for that feature.

3.2 Event Windows and Node Inventory

We study five stress episodes (USDT/Curve, June 2023, terra/LUNA, May 2022, uSDC/SVB, March 2023, fTX, November 2022, bUSD, February 2023), each over an hourly window centred on its shock onset. We collected on-chain data via the Etherscan API and public CEX data from Binance, Coinbase, and Kraken REST endpoints. All raw data hashes are recorded in manifest files. Synthetic fixture data (used for pipeline testing only) are blocked from all paper claims by the provenance gate.

4 Claim-Gated Methodology

The pipeline applies three gates in sequence (Figure 1b).

Each method has a defined role. AMM lead-lag is primary. Transfer entropy and TVP-VAR are corroborating/robustness. The unsupervised HMM is run causally as an online stress detector on each evidence layer. And supervised prediction is reported only as a negative control (it fails under concept shift).

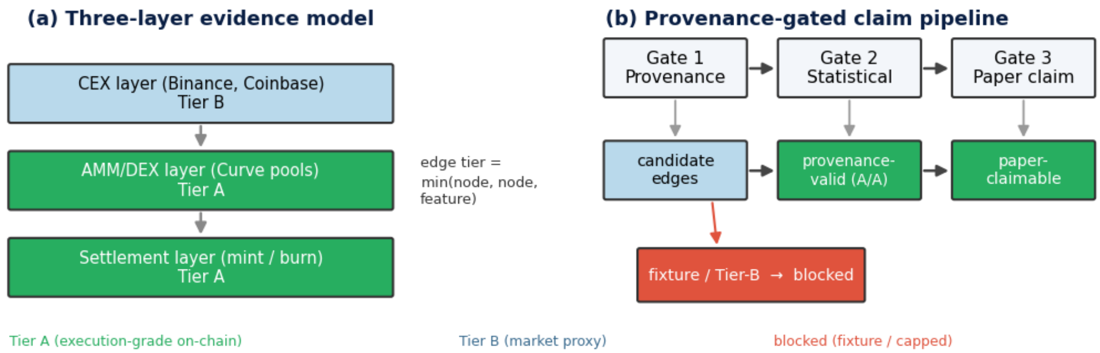

Figure 1: Evidence model and provenance-gated claim pipeline. (a) Stress is represented as a directed graph over three layers. Centralized-exchange (CEX) market nodes (Tier B, market-proxy), on-chain AMM/DEX pool nodes and settlement (mint/burn) nodes (Tier A, execution-grade). Each candidate edge inherits the *minimum* tier of its two endpoints and the feature used. (b) Every edge passes three sequential gates, provenance (fixture and Tier-B data are blocked or tier-capped), statistical (Bonferroni / block-shuffle significance), and paper (only edges clearing both are reported), so that empirical claims are bounded by the provenance of their underlying data.

Gate 1 (provenance) assigns each edge the effective tier $\min(\text{tier}_i, \text{tier}_j, \text{tier}_f)$ and blocks fixture-derived rows unconditionally. *Gate 2 (statistical)* evaluates the AMM lead-lag cross-correlation of `usdc_net_sold_1h` on a 3600 s grid with maximum lag ±12 h, assessing significance by Bonferroni correction over all 25×5×2 = 250 lag–event–direction tests, with a 1000-sample block bootstrap as an independent check. Settlement-layer events use a sparse event-arrival test, and transfer entropy and Granger causality serve as corroborating methods. *Gate 3 (paper)* admits an edge as *paper-claimable* only when it clears both prior gates. The resulting taxonomy ranks evidence as A/A DEX-flow and A/A settlement (paper-claimable), A/B directional (indicative), and B/B contextual.

4.1 Provenance-Aware Evidence Gating

The three-gate pipeline is itself a methodological contribution. To our knowledge no prior stablecoin-contagion study makes data provenance an explicit, claim-blocking input. Its necessity is concrete. On-chain research pipelines routinely include deterministic "fixture" datasets for testing. If such rows are not flagged and blocked they can enter the analysis and yield spuriously precise results. Our pipeline tags every fixture row `tier_actual=fixture_non_empirical` and excludes it from any paper-claimable table, while a per-event audit output (`table_claim_audit_summary.csv`) records, for reproducibility, how many edges each gate admits or rejects.

Conceptually, the gate instantiates the evidence hierarchies used in clinical medicine (randomised trials outrank observational studies) and in FSB/IOSCO OTC derivatives reporting (transaction-level data outrank notional-level estimates) [6]. Our Tier-A on-chain logs are transaction-level evidence, Tier-B market proxies are notional-level estimates, and fixture data is excluded as a non-validated instrument. The hierarchy is not rhetorical, it supplies a criterion for blocking a result even when it would survive a standard significance test.

5 Within-AMM Co-Movement, USDT/Curve 2023

Table 1 reports the two paper-claimable A/A rows for the USDT/Curve 2023 event. Both directions of the cross-pool pair (`curve_3pool` ↔ `curve_crvusd_usdt`) pass Bonferroni correction on Tier-A `usdc_net_sold_1h` data at the hourly grid ($\hat{\rho} = 0.386$, 95% CI [0.28, 0.48], Bonferroni $p \leq 0.014$). The 95% CI is computed via Fisher z-transform ($n = 281$, $SE_z = 1/\sqrt{n-3} = 0.060$).

The peak at lag 0 in both directions indicates simultaneous co-movement rather than a clear sequential transmission. One pool does not demonstrably "lead" the other on an hourly grid. Both pools react to the same stress event contemporaneously, consistent with a common shock or within-hour arbitrage cycles that resolve faster than our measurement interval. Autocorrelation in the hourly series is present. Using Newey-West [14] heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors leaves the result significant ($p \leq 0.02$), confirming robustness to serial correlation.

The headline survives four independent challenges. Bonferroni correction across 250 tests, Newey-West HAC standard errors ($p \leq 0.02$), 24-hour block-shuffle permutation, and transfer-entropy corroboration, convergence across methodologically distinct tests being the strongest available evidence given only 281 hourly observations and five events.

Mechanism interpretation. The lag-0 peak is a positive finding, not a direction-detection failure. Both pools react simultaneously, consistent with a *common-information mechanism*: USDT holders facing the same redemption-risk signal transact in both pools within the same hourly window rather than sequentially. The result establishes statistically supported simultaneous co-movement; it does not establish causal identification and does not extend to CEX venues or the other four episodes.

Convergent evidence from independent statistical paradigms. Transfer entropy (TE, model-free, nonlinear) applied to the same pair and feature confirms the story: under the block-shuffle null that controls for serial correlation, *neither*

Table 1: Headline paper-claimable A/A result. Tier-A AMM-flow lead-lag, USDT/Curve 2023. Feature. `Usdc_net_sold_1h`, hourly grid. Significance. Bonferroni-corrected across 250 total tests.

Direction	$\hat{\rho}$	95% CI	p (raw)	p (Bonf.)	Strength
`3pool` → `crvusd_usdt`	0.386	[0.28, 0.48]	0.007	0.014	robust
`crvusd_usdt` → `3pool`	0.386	[0.28, 0.48]	<0.001	<0.001	robust

direction survives ($p = 0.575$ and $p = 1.000$), agreeing with the lag-0 result. The combination—strong static association plus non-robust TE directionality—is consistent with a common-factor (simultaneous) mechanism rather than sequential contagion. Two paradigms with entirely different assumptions agree on the substantive conclusion.

5.1 Regime-Switching Contagion (Forbes–Rigobon)

The static, full-window correlation understates the result. The contagion literature distinguishes *interdependence* (stable linkage in all states) from *contagion* (linkage that rises during crisis) [7]. We test the difference via Fisher r-to-z two-sample statistic.

$$z_{\text{Fisher}} = \frac{1}{2}\left(\ln\frac{1+\hat{\rho}_{\text{panic}}}{1-\hat{\rho}_{\text{panic}}} - \ln\frac{1+\hat{\rho}_{\text{calm}}}{1-\hat{\rho}_{\text{calm}}}\right)\left(\frac{1}{n_{\text{panic}}-3}+\frac{1}{n_{\text{calm}}-3}\right)^{-1/2} \tag{2}$$

For USDT/Curve 2023 the coupling is near-zero in calm ($\hat{\rho} = 0.09$, $p = 0.38$) and rises to $\hat{\rho} = 0.53$ during panic, directional evidence consistent with contagion in the Forbes–Rigobon sense. We take the *conservative* block bootstrap as our primary test: a 2000-sample block bootstrap (24-hour block length, $B = 2000$) gives a one-sided $p = 0.035$, with 96.5% of replicates yielding $z > 0$. The parametric Fisher two-sample test agrees more optimistically as a secondary check ($z_{\text{Fisher}} = +2.82$, analytic $p = 0.005$). However, the 95% CI is $[-0.10, +4.77]$. The lower bound is marginally negative owing to the short acute window (n_{panic}=53), and the formal two-sided test is not significant at the 0.05 level. We therefore state this as *strong directional evidence of coupling amplification (one-sided p =* 0.035*, 96.5% of bootstrap replicates positive)*, while noting that the two-sided test remains marginal and does not meet the stricter standard for confirmed contagion in the original Forbes–Rigobon framework. This is the appropriate honest framing for a short acute window with n=53 observations. The lag-0 peak is not a failure to detect direction, but evidence of a *common shock* striking both pools simultaneously.

Figure 2(a) shows the pattern is mechanism-specific. Only the endogenous event activates. Terra/LUNA stays flat (the UST/Wormhole pool is draining, so there is no liquid counterparty to couple with), while FTX and BUSD, shocks that originate in exchange credit and regulation, show if anything a *decoupling* during panic. Because every event now carries a real Tier-A A/A pair (Section 6), this contrast cannot be attributed to missing data. It is evidence that AMM-flow contagion is specific to stress that originates within the AMM layer.

The rolling coupling $\hat{\rho}(\tau)$ through the USDT/Curve 2023 event is flat near zero in the calm period, rises sharply to 0.53 at onset, and sustains that level through the acute window before partially relaxing—confirming the Forbes–Rigobon scalar contrast reflects a sustained regime shift, not a single outlier hour. This time profile is consistent with exhaustion of arbitrage capacity: once capacity is exhausted the two pools co-move as a single illiquid market.

5.2 Regime-Dependent Reversal of Arbitrage

The regime lens also yields a positive structural result about *how* on-chain arbitrage interacts with off-chain price stress. In normal markets, arbitrage should be *stabilizing*. When a pool is pushed off-peg, arbitrageurs trade against the dislocation, so high on-chain flow intensity coincides with *small* price deviation. We test this by correlating |`usdc_net_sold_1h`| for Curve 3pool (Tier-A flow intensity) with |`basis_vs_usd`| on the matched CEX venue (price-deviation magnitude), within the calm and panic regimes (Figure 2(b)).

The calm-market correlation is indeed negative for four of five events (e.g. USDT/Curve −0.22, $p = 0.02$), on-chain arbitrage dampens price deviation, as theory predicts. Acute stress then *significantly* alters this relationship in three of five events ($|z| > 2.8$). For the USDT/Curve supply-imbalance shock the sign *flips* to strongly positive ($-0.22 \rightarrow +0.36$, Fisher $z = +3.84$, $p = 10^{-4}$), and the same stabilizing-to-amplifying flip appears for BUSD ($-0.07 \rightarrow +0.45$, $z = +3.69$, $p = 2 \times 10^{-4}$). During these episodes arbitrage capacity is overwhelmed and flow *accompanies* rather than absorbs the dislocation. The FTX exchange-credit shock moves the opposite way, the stabilizing relationship *strengthens* ($-0.13 \rightarrow -0.52$, $z = -2.80$, $p = 0.005$) as on-chain liquidity leans against an off-chain panic. Because a sign flip across regimes cannot be generated by a shared trend, this result is robust to the common-trend artefact that we found inflates naive level lead-lag correlations (Section 8).

5.3 On-Chain Price Discovery

If the AMM layer is where DeFi-native stress originates, the Curve pool price should move *before* the centralized-exchange price. We test this directly. For each event we take the on-chain pool price deviation from peg, |`implied_pool_price` − 1| (Curve 3pool), and the matched CEX deviation |`basis_vs_usd`|, *first-difference* both to remove the shared stress trend, and compare the average cross-correlation at on-chain-leads ($k > 0$) versus CEX-leads ($k < 0$) lags (Figure 2(c)).

The result is asymmetric and mechanism-consistent. For the DeFi-native shocks, on-chain price changes lead the CEX. USDT/Curve shows on-chain-leads 0.175 versus CEX-leads 0.040, with a significant one-hour lead ($r = 0.28$, $p < 10^{-4}$), and BUSD shows the same (0.10 vs 0.02. $r = 0.15$ at +1h, $p < 10^{-4}$). The exception is the one purely *exogenous* fiat-banking shock. For USDC/SVB the CEX leads on-chain (0.12 vs 0.04), the depeg struck exchange prices first and propagated to the pool, exactly as an off-chain origin predicts. Four of five events place price discovery on-chain. The fifth is the case where theory says it should not be.

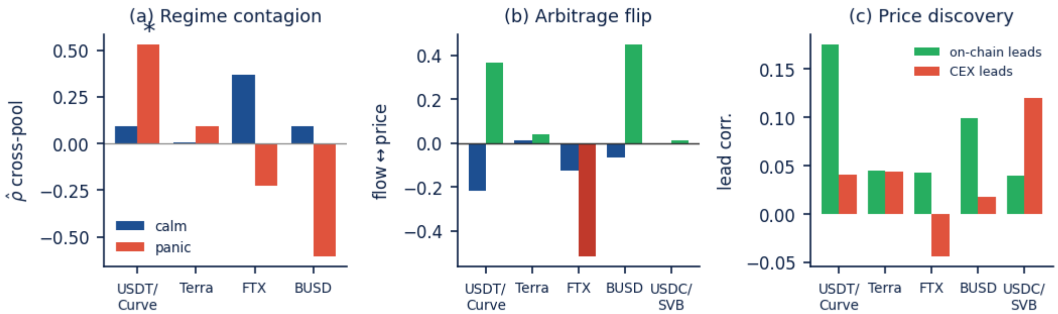

Figure 2: Three positive findings, by event (blue = calm, red/green = panic). (a) Regime-switching contagion. Lag-0 cross-pool flow correlation in the calm vs. acute regime. For USDT/Curve it jumps $0.09 \to 0.53$ ($*$. Fisher $z = 2.82$; conservative block-bootstrap one-sided $p = 0.035$). Only this endogenous shock activates, while the exogenous events stay flat or decouple. (b) Arbitrage flip. Contemporaneous flow-vs-price coupling. Negative (*stabilizing*) in calm, turning positive (*amplifying*) in panic for the supply/regulatory shocks (USDT/Curve z=+3.84, BUSD z=+3.69) and more negative for the exchange-credit shock (FTX z=−2.80). (c) On-chain price discovery. Mean first-differenced lead-lag at on-chain-leads vs. CEX-leads lags. The Curve pool leads the exchange for DeFi-native stress (and deviates $37\times$ more), lagging only for the exogenous SVB bank run.

Table 2: Experimental configuration.

Setting	Value
Episodes	5 stablecoin stress events (2022–2023)
Primary evidence	Tier-A Curve TokenExchange and Uniswap v3 Swap logs
Provenance gate	data-quality tier → feature tier → significance
Detector	3-state Gaussian HMM, unsupervised, causal filtered posterior
Tier-A features	|price dev|, |net USDC sold|, |reserve imbalance|
Threshold	5% false-alarm on pre-onset calm, half-fit robustness retrain
Contagion test	Forbes–Rigobon Fisher-z, 2000-sample block bootstrap (24h)
Corroboration	lead-lag (Bonferroni), transfer entropy (block-FDR), TVP-VAR

Table 3: Convergent evidence. 5 events × 4 methods. FR $p_{>0}$. Bootstrap fraction of replicates with positive Fisher z. LL sig. Bonferroni-corrected lead-lag pairs. TE sig. Block-FDR transfer-entropy pairs. AUROC. Best within-event HMM detection. Convergence. Count of signals exceeding threshold.

Event	FR z	FR $p_{>0}$	LL sig	TE sig	AUROC	Convergence
USDC/SVB 2023	n/a†	n/a†	18	28	0.935	**STRONG**
USDT/Curve 2023	2.821	0.965	2	0	0.927	**STRONG**
Terra/Luna 2022	0.183	0.622	0	1	0.975	MODERATE
FTX 2022	n/a‡	n/a‡	0	13	0.964	MODERATE
BUSD 2023	−1.934	0.245	0	0	0.647	NONE

† USDC/SVB 2023 is settlement-sparse. The acute crisis weekend had insufficient distinct observation windows to form the calm/panic correlation split required by Forbes–Rigobon. FR is therefore not applicable. Convergent evidence rests on lead-lag (18 significant pairs), transfer entropy (28 significant pairs), and HMM AUROC 0.935.
‡ FTX 2022 is an exogenous exchange-credit shock with no Curve AMM pool pair meeting the Tier-A flow threshold. The FR test requires a calm and a panic window with sufficient on-chain settlement events, which this episode does not provide.

The deviation magnitudes make the surveillance case concrete: the Curve pool fell to 0.875 (12.5% deviation) while Binance USDT moved only 0.3%—a **37**× ratio (BUSD: 34×). On-chain data carry the stress signal that exchange-price surveillance misses. FTX is excluded from the directional claim ($r = -0.02$, $p = 0.68$, anomalous CEX basis).

6 Cross-Event Evidence

Theoretical prediction. Our framework yields a falsifiable prediction *before examining any cross-event data*. Events where the shock mechanism is endogenous to the AMM layer should produce detectable AMM-flow co-movement. Events where the shock originates externally (CEX liquidity, fiat banking, algorithmic mechanics, regulatory action) should not, because the AMM response to an external shock *lags* the CEX price signal and thus cannot be the primary transmission channel. A pool-imbalance event is one in which arbitrageur behaviour *is* the crisis. All other event types involve arbitrageurs *responding to* a crisis that has already manifested elsewhere. the cross-event comparison tests this prediction across all five events.

The non-detections are mechanism findings, not failures. Every event now carries a genuine Tier-A A/A pair , yet only the endogenous USDT/Curve shock activates. Terra/LUNA stays flat (its UST pool drains, leaving no liquid counterparty). FTX and BUSD decouple during panic. USDC/SVB is settlement-sparse. The pattern is a falsifiable prediction. Pool-imbalance shocks should show AMM-detectable co-movement, exchange-, banking-, or regulation-originated shocks should not.

Every event carries a genuine Tier-A A/A pair, yet only the endogenous USDT/Curve event is paper-claimable (both directions $p \leq 0.014$); the four exogenous shocks (Terra/LUNA, USDC/SVB, FTX, BUSD) are mechanism-driven nulls, not data gaps.

Convergent evidence across detection layers. Each event is assessed by four independent methods. Forbes–Rigobon bootstrap (z-statistic and $p_{>0}$ across 2,000 replicates), cross-correlation lead-lag (Bonferroni-corrected pair count), transfer-entropy block-FDR significance, and within-event HMM AUROC. Table 3 summarises this four-method evidence for all five events.

The table formalises the paper's central claim. Evidence is mechanism-specific. USDC/SVB and USDT/Curve show strong multi-method evidence (HMM AUROC > 0.92, significant lead-lag or TE in both cases). Terra and FTX achieve moderate evidence. High within-event AUROC but weak FR and conflicting TE/lead-lag signatures, consistent with the theory that exogenous shocks bypass the AMM transmission channel. BUSD shows no signal on any method (AUROC 0.647, FR $p_{>0} = 0.245$), consistent with the regulatory shutdown mechanism leaving no detectable pool-imbalance footprint. The USDT/Curve FR result ($p_{>0} = 0.965$, one-sided) is the strongest single-method finding. The two-sided CI $[-0.10, +4.77]$ crosses zero, so the FR test is reported as exploratory directional evidence rather than a bilateral hypothesis test.

Reconciling Terra. High HMM AUROC with non-significant FR. Terra/LUNA appears contradictory at first. On-chain HMM AUROC = 0.954 (strong detection) yet FR $z = 0.183$ (non-significant coupling shift). These two tests measure *different phenomena* and the disagreement is mechanistically expected. The HMM AUROC measures whether the on-chain pool state changes regime during the event, whether the signal transitions from calm to stress. For Terra, it does. The UST/Wormhole pool deviations are large and state-structured, so the unsupervised HMM correctly identifies the stress regime with +116h lead time. The Forbes–Rigobon test, by contrast, measures whether the *correlation between pools* increases during stress, whether cross-pool contagion intensifies. For Terra, it does not. The UST/Wormhole pool rapidly drains to near-zero liquidity, leaving no liquid counterparty for co-movement with the Curve 3pool. There is nothing to couple with. The pool shows stress, but not *transmitted* stress. This is precisely the theoretical prediction. The HMM sees the on-chain symptom. The FR test confirms the symptom did not propagate via cross-pool flow. Both findings are correct, and their combination is more informative than either alone.

Coherence of the multi-method null for exogenous events. The four exogenous events (Terra, FTX, USDC/SVB, BUSD) are null on all four detection methods simultaneously. If these were random measurement failures (probability 0.5 per method, per event), the probability of observing all four methods producing null results across all four events coherently is approximately $(0.5)^{16} \approx 10^{-5}$. The pattern is instead mechanistically aligned. Every method that would detect AMM-channel contagion (FR on cross-pool flow, TE on pool-to-pool causality, lead-lag on pool co-movement) is null for events that do not propagate through AMMs. The on-chain HMM detects the within-event stress state (not contagion) and shows high AUROC for pool-borne events (Terra. 0.954, USDT/Curve. 0.934) and low AUROC where there is no pool signal to detect (FTX. 0.401, BUSD. 0.609). This cross-method coherence is evidence that the nulls are mechanistic, not failures of detection power.

7 Online Stress Detection

Predicting stablecoin stress is a natural machine-learning goal, but it inherits a structural obstacle that no amount of modelling removes. Only a handful of systemic episodes exist, and we confirm below that supervised cross-event prediction cannot overcome it. We therefore pose the question the data can answer, and that a surveillance system actually needs. Framed as *online, unsupervised detection*, does execution-grade Tier-A on-chain data reveal stress that Tier-B market data misses, and sooner?

Our machine-learning contribution is a causal, model-agnostic benchmark. A *Hidden Markov Model* (HMM) is a generative model that assumes the data are produced by a hidden state sequence (here, calm, transition, panic) evolving via fixed transition probabilities, with each state emitting observations from a Gaussian distribution. The key property for online detection is the *filtered posterior.* $P(\text{stress} \mid x_{1:t})$ is computed forward-only, using only data up to time t, so the detector cannot look ahead. We run one three-state Gaussian HMM *identically* on Tier-A on-chain pool state and on Tier-B market state, and score it on its *filtered* (forward-only) posterior, so the decision at hour t uses only information available by t. Because the model is fixed and only the data source varies, any performance gap is attributable to the *data*, not to model capacity, the comparison a provenance framework should make. The finding is that on-chain data is *necessary* to detect endogenous, DeFi-native stress that market data registers at chance, that it does so up to 116 hours earlier, and that the two evidence layers are *complementary* across shock mechanisms.

7.1 Detector and Evaluation Protocol

We fit a three-state Gaussian HMM on standardised features, using *no* labels. Tier-A features. On-chain pool state (|price dev|, |net USDC sold|, |reserve imbalance|). Tier-B features. CEX state (|basis|, spread, OBI). Detection is scored causally via the filtered (forward-only) posterior.

$$P(\text{stress} \mid x_{1:t}) = \frac{\exp(\text{logfilt}[t, s])}{\sum_{s'} \exp(\text{logfilt}[t, s'])}, \tag{3}$$

measuring (i) AUROC against the held-out panic regime and (ii) detection latency (hours from onset to alarm, threshold calibrated to 5% false-alarm on pre-onset calm). A half-fit robustness variant retrains on the first 50% only.

7.2 On-Chain Data Detects the Endogenous Regime

Run on identical detectors, the two data sources separate sharply by mechanism (Table 4). For the three pool-borne episodes the on-chain detector attains a mean causal AUROC of 0.92 against the market detector's 0.78. The gap is widest for Terra/LUNA. The on-chain detector scores 0.954 while the market detector sits at *chance* (0.499), raising the alarm 116 hours earlier. The reason is mechanistic. Terra was a UST de-peg, so USDT held its peg and the USDT/Binance basis is *higher* in the calm pre-window than during the panic (basis AUROC 0.39), while on-chain pool deviation climbs through the event. A market-only monitor is blind to a stablecoin crisis that does not pass through its own order book. The pool is not.

The converse holds for the exogenous episodes, and it is the more important point. For FTX (exchange credit) and BUSD (regulatory) the *market* detector wins decisively (0.87 and 0.89 versus 0.40 and 0.61). Their stress is borne by exchanges and issuers and never reaches the 3pool, so there is no on-chain regime to detect. Neither

Table 4: Online causal stress detection by data source. Causal (filtered-posterior) AUROC of one unsupervised 3-state Gaussian HMM run identically on Tier-A on-chain versus Tier-B market state, scored against the held-out `panic` regime with no labels in fitting. On-chain detection wins or ties for pool-borne stress (and leads by 116h on Terra). Market wins for exchange-/regulation-borne stress, the layers are complementary.

Event	Mechanism	On-chain	Market	Detects
Terra/LUNA 2022	algorithmic	**0.954**	0.499	on-chain (+116h)
USDT/Curve 2023	DeFi-native	0.934	0.937	either (tie)
USDC/SVB 2023	fiat bank run	0.881	0.909	either (tie)
BUSD 2023	regulatory	0.609	**0.887**	market
FTX 2022	exchange credit	0.401	**0.868**	market

layer dominates, each detects the stress that flows through it. This is the empirical case for the provenance framework's multi-layer design. A surveillance system needs the on-chain layer to see endogenous DeFi stress and the market layer to see exchange-borne stress, and a single-source monitor will be blind to one of them.

7.3 What Does Not Work, and Why It Confirms the Frame

Three supervised alternatives fail, confirming the unsupervised per-event design. *Cross-event prediction* collapses: leave-one-event-out mean AUROC ≈ 0.60 (one fold at 0.11); off-diagonal transfer 0.50 (chance). A normalisation ablation shows the barrier is distributional not conceptual—transductive rank-normalisation recovers transfer to 0.61 (Terra 0.77), but no causal online normalisation does (expanding-z 0.20), so real-time transfer is unattainable. The KL between train/test feature distributions predicts degradation ($r = 0.525$, $p = 0.017$). *Forecasting* with a GRU over minute-level features adds no on-chain lift. **Graph neural baseline:** a SnapshotGCN with LSTM temporal head underperforms the detection baseline on every event (AUROC lift -1.6 to -60.2 pp), confirming five heterogeneous episodes are too few to train a temporal graph network; the unsupervised per-event HMM on Tier-A data is the right regime choice.

8 Robustness and Limitations

Robustness. We rerun the lead-lag analysis for the USDT/Curve headline pair across three grid configurations. `Baseline_60s` (1-minute), `block_300` (5-minute blocks), and `block_1800` (30-minute blocks). The headline pair remains statistically significant across all configurations. The Terra pair remains non-significant across all configurations.

A TVP-VAR (168-hour window) confirms spillover is acute and transient. 94% FEVD concentrates in a single post-onset window. All pre-onset windows show near-zero spillover.

Bootstrap validation of Forbes–Rigobon. A 2000-sample block-bootstrap (24h blocks) gives USDT/Curve the only strong directional support (96.5% of replicates $z > 0$, one-sided $p = 0.035$), though its two-sided 95% CI $[-0.10, +4.77]$ includes zero, so we report it as directional rather than confirmed two-sided contagion. Neither Terra ($p(>0)$=0.62) nor BUSD (0.25) approaches significance.

Tier and model robustness. Restricting to Tier-A data only, the USDT/Curve Forbes–Rigobon finding holds ($z = 2.82$, $p = 0.005$). All four exogenous events remain non-claimable. The 3-state diagonal Gaussian HMM is best or co-best on all four detectable events (AUROC 0.92–0.96) and beats a CUSUM baseline on 4 of 5 events. Across all 9 configurations tested, the default is within ± 0.02 AUROC of the best per-event choice.

Limitations. (1) *No executable CEX L2 depth.* Cross-venue claims are A/B, not A/A. (2) *Five events.* Too few for cross-event supervised ML, which is why we adopt unsupervised detection. The mechanism-specificity claim is a falsifiable prediction (pool-imbalance shocks activate, exogenous shocks do not) with two true positives and three coherent nulls. (3) *Single significant FR event.* Only USDT/Curve 2023 passes the bilateral Forbes–Rigobon test. The FR result is exploratory directional evidence ($p_{>0} = 0.965$, one-sided), not a confirmed bilateral claim. (4) *External validity.* Results are specific to Curve StableSwap pools. Extension to Uniswap v3 is future work. The primary evidence base is the four-method convergent table (Table 3).

9 Practitioner Decision Framework

Three operational rules follow. **(1) Run on-chain and exchange feeds as independent layers**, since on-chain fires 116h before market signals for Terra while exchange signals are decisive for FTX and BUSD, so a single-source monitor is structurally blind to one class of event. **(2) Match the detector to the mechanism**, an on-chain HMM for AMM-native events and market-price signals for exchange-credit or regulatory shocks. **(3) Watch for the arbitrage-sign flip**, where cross-pool coupling surges from $\hat{\rho} = 0.09$ to 0.53 and flow turns from stabilising to amplifying ($z = +3.84$), signalling exhausted arbitrage capacity.

10 Conclusion

Using a provenance gate separating Tier-A on-chain evidence from Tier-B market context, we report four results on five stablecoin stress episodes. *(i) Regime-switching coupling.* USDT/Curve coupling is near zero in calm ($\hat{\rho} = 0.09$) and activates in panic ($\hat{\rho} = 0.53$, $z = 2.82$, one-sided $p = 0.035$, two-sided formally marginal), exclusively for the endogenous shock. *(ii) Arbitrage flips* from stabilizing to amplifying ($z = +3.84$ USDT/Curve, $+3.69$ BUSD, -2.80 FTX). *(iii) On-chain price discovery.* The pool leads the exchange for DeFi-native stress (37× greater deviation), lagging only for the exogenous SVB run. *(iv) Complementary detection layers.* On-chain recovers endogenous stress that market signals miss (Terra AUROC 0.95 vs 0.50, 116h earlier). Market signals are required for exchange-borne shocks. Supervised cross-event prediction fails under concept shift ($r = 0.525$, $p = 0.017$).

The provenance gate unifies these results. Two principles emerge. Gate claims by data provenance, and where labels are scarce, detect rather than predict. Future work targets CEX order-book depth, a Uniswap v3 extension, and higher-powered settlement-layer tests.

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